

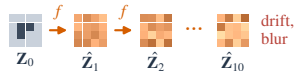
### (a) Move, don't regenerate



observed patches  $Z_0$

+ actions  $\rightarrow$  forecast  $\hat{Z}_{1:10}$

Existing WMs: regenerate every patch



ShiftWM: move what it has seen



$$\hat{Z}_k = (1-g)Z_0 \text{ keep} \\ + gT_k \text{ move} \\ + r_k \text{ correct}$$

all horizons in parallel from measured  $Z_0$

### (b) One held-out DROID window

true  $t+10$



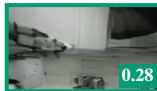
Direct



AR-TF



ShiftWM



AR



Persistence

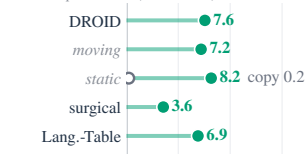


corner: feature error at  $t+10$  ( $\downarrow$ )

### (c) Held-out error reduction

lower error than Direct in 100/100 DROID horizon $\times$ motion bins

matched predictors (same data)



plug-in head (vs. model w/o head)

